

ELEC 451 Power Electronics

Lab 1: Power Quality and Basic Uncontrolled Rectifiers

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I. INTRODUCTION

II. SIMULATION

A. Circuit 1 — Full-Bridge Rectifier with Resistive Load

B. Circuit 2 — Full-Bridge Rectifier with Capacitive Filter and Resistive Load

C. Circuit 3 — Full-Bridge Rectifier with LC Filter and Resistive Load

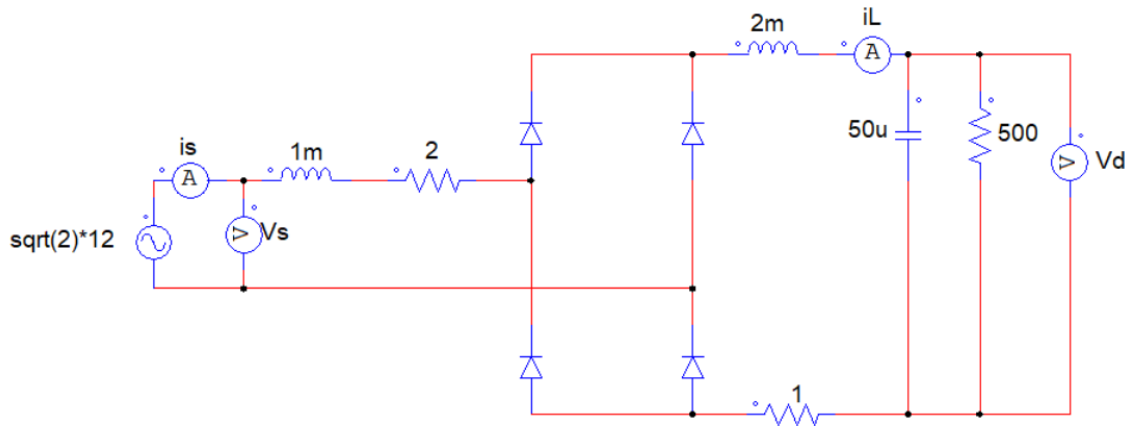


Fig. 1. PSIM schematic of Circuit 3.

The THD can be calculated as (these are the source voltage and current):

$$\text{THD} = \frac{\sqrt{\sum_{h \neq 1} I_{sh}^2}}{I_{s1}} \quad (1)$$

equivalently...

$$I_s = I_{s1} \sqrt{1 + \text{THD}^2} \quad (2)$$

$$\frac{I_s^2}{I_{s1}^2} - 1 = \text{THD}^2 \quad (3)$$

where I_{s1} is the fundamental frequency of the source current.

We know at steady state, the $I_s = I_L = I_o$. And since the load determines how much current flows through from the source,

$$I_{s1} = \frac{V_s}{R + R_{\text{shunt}} + R_s} \quad (4)$$

$$I_{s1} = \frac{12 \text{ V}}{500 + 1 + 2} \quad (5)$$

$$\boxed{I_{s1} = 23.86 \text{ mA}} \quad (6)$$

This is the fundamental component of $I_{s,\text{RMS}}$ since the harmonic distortion comes from the current ripple.

Method: phasor analysis is used to find the output current, which is then converted to the time domain. Since the output is rectified, the final waveform is restricted to be positive, $i_o(t) = |I_o| |\sin(\omega t + \angle I_o)|$.

As shown in Fig. 1, the source voltage after the source inductance and resistance drop is

$$V_{s2} = V_s - (j\omega L_s + R_s)I_s \quad (7)$$

The output impedance seen by V_{s2} is

$$Z_o = \left(\frac{1}{j\omega C} \parallel R \right) + j\omega L + R_{\text{shunt}} \quad (8)$$

At steady state $i_s = i_L = i_o$, so combining the equations gives

$$I_o Z_o = V_{s2} \quad (9)$$

$$I_o Z_o = V_s - I_o(j\omega L_s + R_s) \quad (10)$$

Solving for I_o with the circuit values,

$$I_o = \frac{12}{0.002j\omega + 503} \quad (11)$$

Converting to the time domain, with consideration for the rectified output,

$$i_o(t) = 23.86 \text{ mA } |\sin(120\pi t - 0.001498)| \quad (12)$$

Note: $|I_o|$ equals I_{s1} calculated earlier, which is a good sign.

Taking the Fourier series, the harmonic content can be analyzed as

$$|\sin x| = \frac{2}{\pi} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\cos(2nx)}{4n^2 - 1} \quad (13)$$

$$|\sin(120\pi t)| = \frac{2}{\pi} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\cos(240\pi nt)}{4n^2 - 1} \quad (14)$$

Since i_o and v_{s2} are essentially in phase, the harmonics can be approximated without accounting for the phase shift.

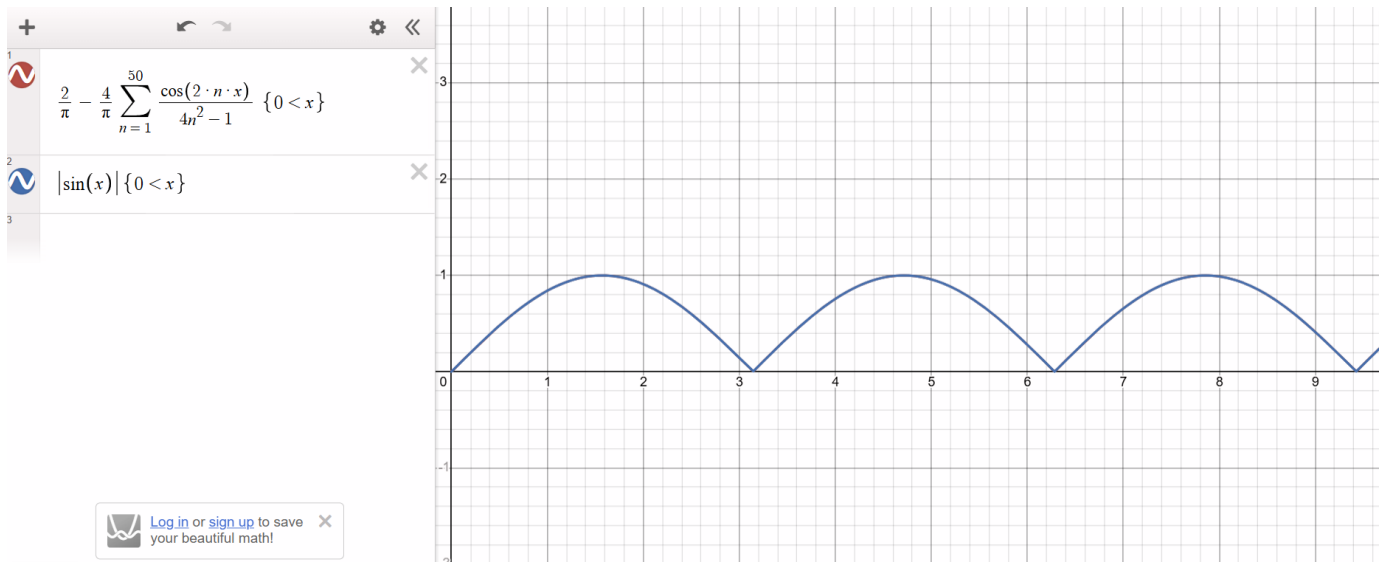


Fig. 2. Fourier-series reconstruction of $|\sin x|$ using the first 50 harmonics.

The peak of each harmonic (when the cosine equals one), ignoring the current amplitude, is

$$I_{p,n} = \frac{4}{\pi(4n^2 - 1)} \quad (15)$$

The fundamental ($f = 120 \text{ Hz}$, $n = 1$) provides the most power, so the power contribution is compared as

$$\frac{I_{p,n}^2}{I_{p,1}^2} \times 100\% \quad (16)$$

where $I_{p,1} = 4/(3\pi)$.

TABLE I
HARMONIC PEAK AMPLITUDES AND POWER CONTRIBUTIONS OF THE RECTIFIED WAVEFORM.

n	$I_{p,n}$	% Contribution
1	$4/(3\pi)$	100%
2	$4/(15\pi)$	4.00%
3	$4/(35\pi)$	0.73%
4	$4/(63\pi)$	0.23%
5	$4/(99\pi)$	0.092%

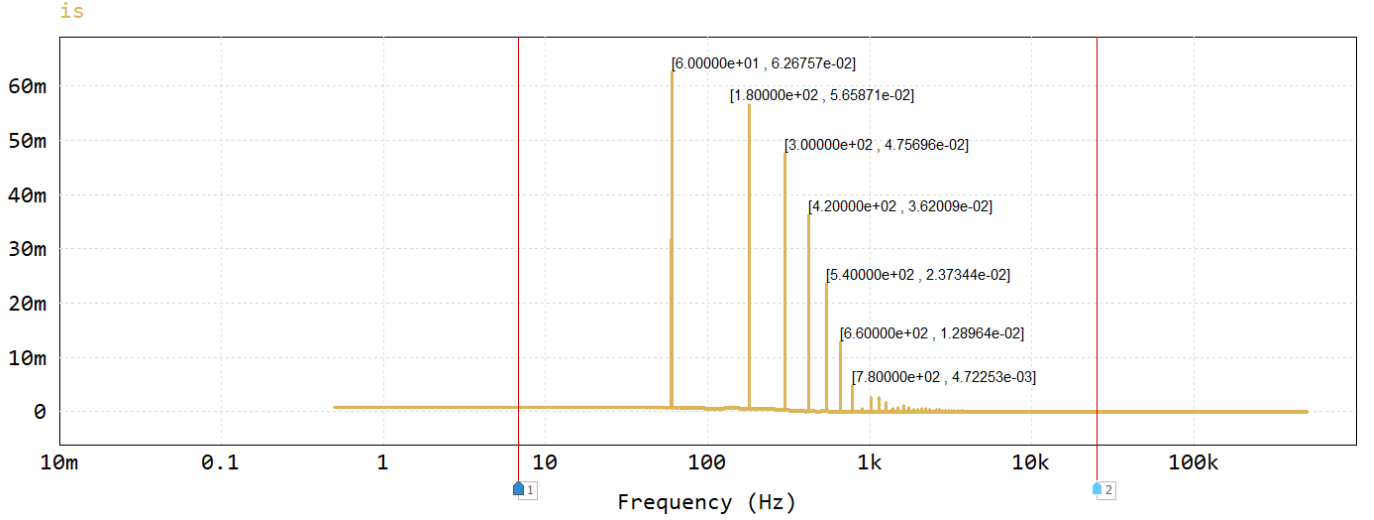


Fig. 3. PSIM FFT simulation of Circuit 3.

TABLE II
CIRCUIT 3 SOURCE-CURRENT FREQUENCY HARMONICS.

f (Hz)	Amplitude (mA)
60	62.68
180	56.59
300	47.57
420	36.20
540	23.73
660	12.90
780	4.723

The fundamental is 60 Hz, and the THD is computed from the amplitudes in Table II:

$$\text{THD} = \frac{\sqrt{(56.59^2 + 47.57^2 + 36.20^2 + 23.73^2 + 12.90^2 + 4.723^2)} \text{ mA}}{62.68 \text{ mA}} \quad (17)$$

$$\text{THD} = \frac{86.76 \text{ mA}}{62.68 \text{ mA}} = 1.384 \quad (18)$$

$$\boxed{\text{THD} = 138.4\%} \quad (19)$$

III. EXPERIMENT

- A. Circuit 1 — Resistive Load
- B. Circuit 2 — Capacitive Filter and Resistive Load
- C. Circuit 3 — LC Filter and Resistive Load

IV. CONCLUSION